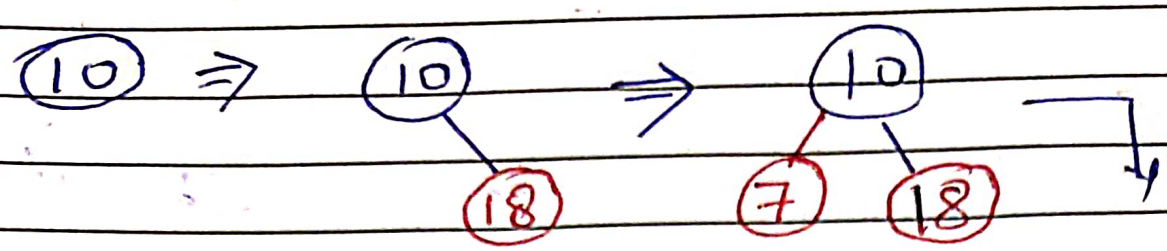
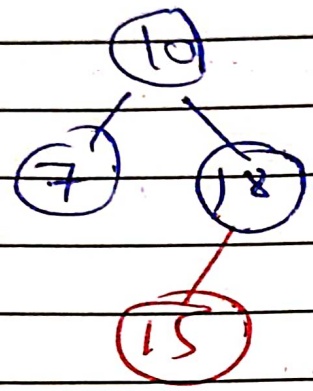
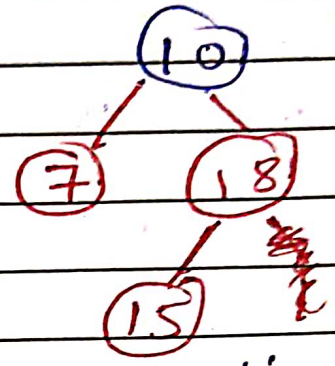
→ root = black
→ no two adjacent red nodes
→ count of black nodes should be same in each path.

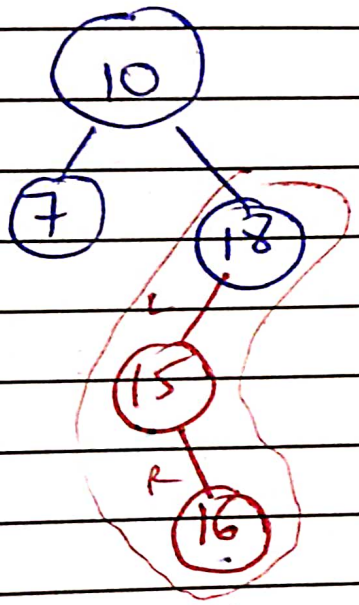
10, 18, 7, 15, 16, 30, 25, 40, 60, 2, 1, 70





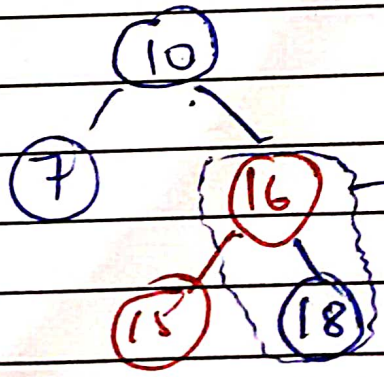


* No two adjacent can be red
* parent's sibling is Red (i.e. 15th parent's 18 & 18th sibling is 17th) then recolor parent & parent's sibling
* Here 15th parent's (18) parent (10) is root ∴ no further checking

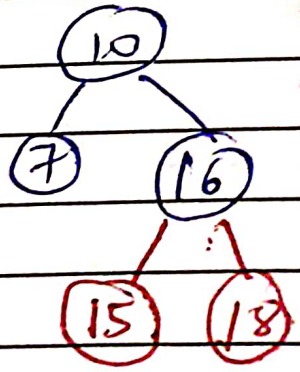


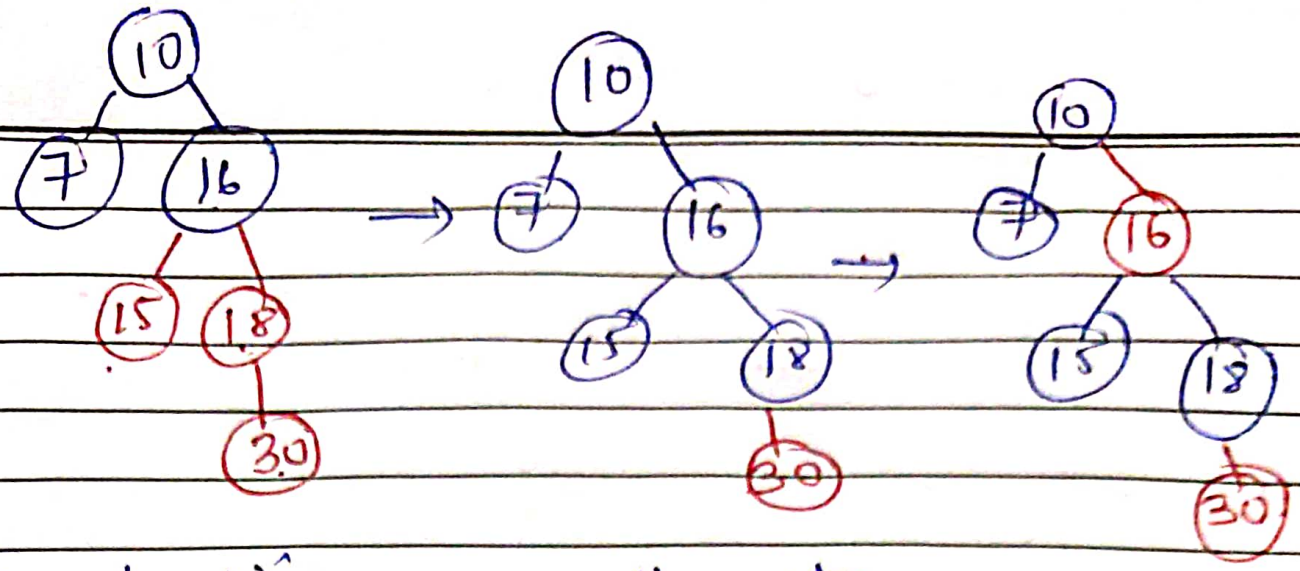
* parent's sibling is Null
rotation & recolor

after the rotation rgb two nodes will be considered for recoloring



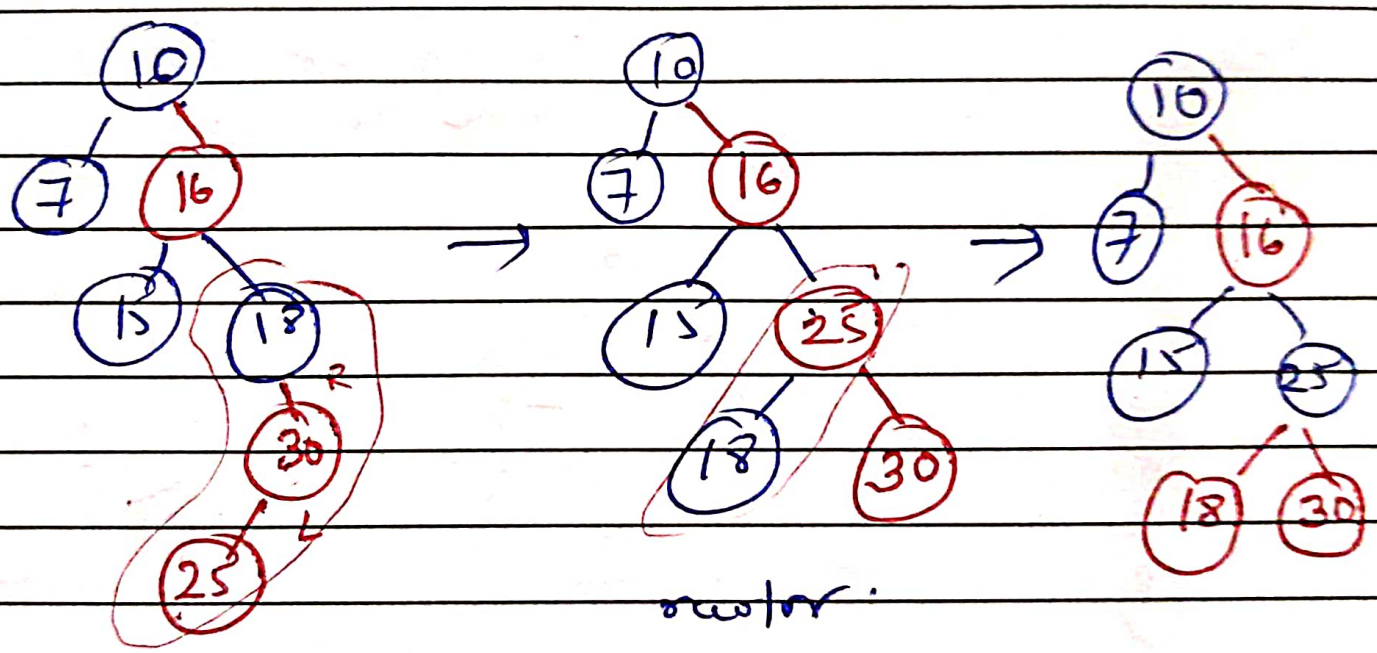
recolor (16)





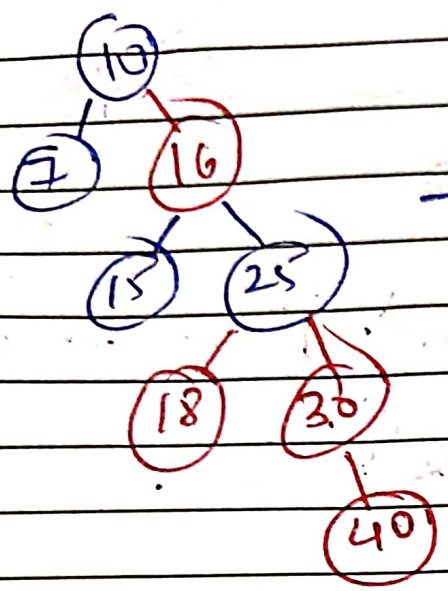
* parent's sibling is red
recolor 18 & 15

* parent's parent is not root node (recolor)

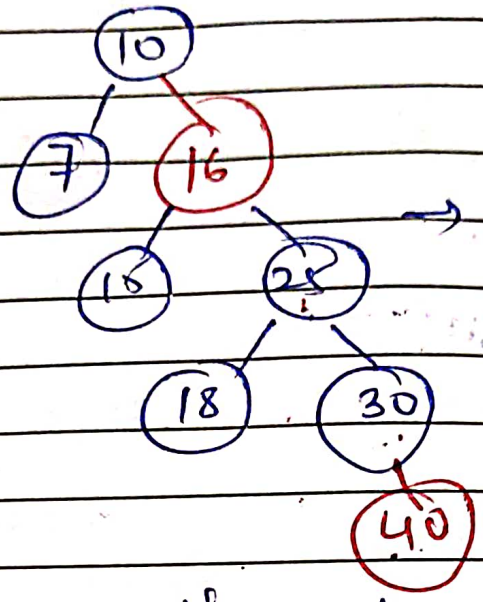


* parent's sibling is null
rotation & recolor

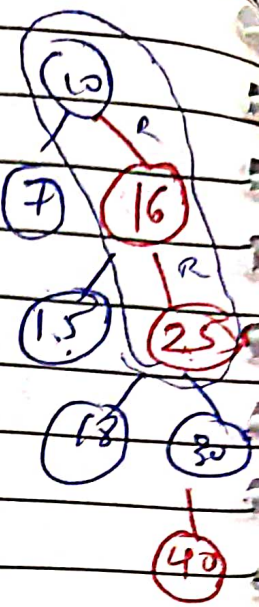
insert 40



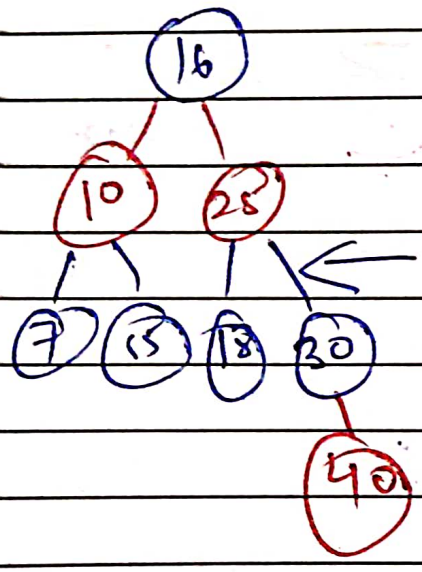
parent's sibling is red, then recolor.

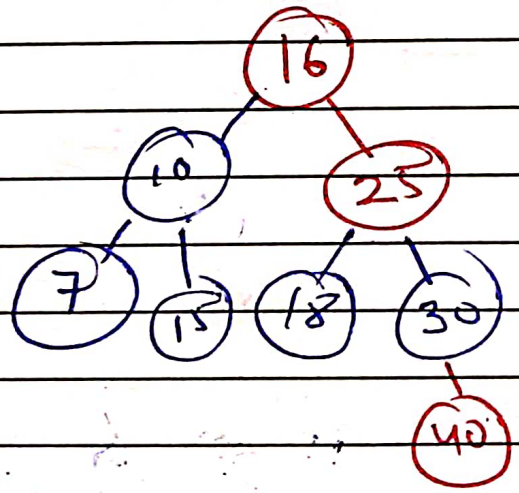


parent's parent is not root node then recolor.



but here after using 25, violates the properties of RB tree





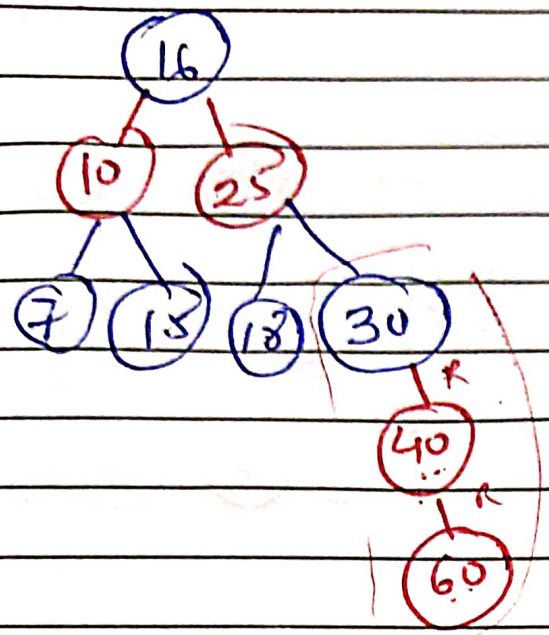
now 25 become new node

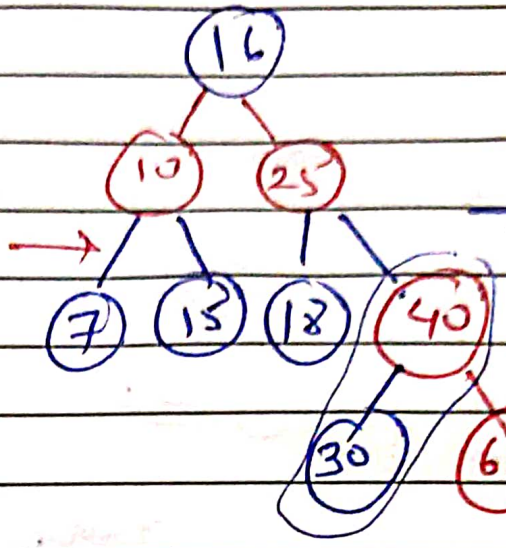
parent's sibling is black (16) (7) then rotation.

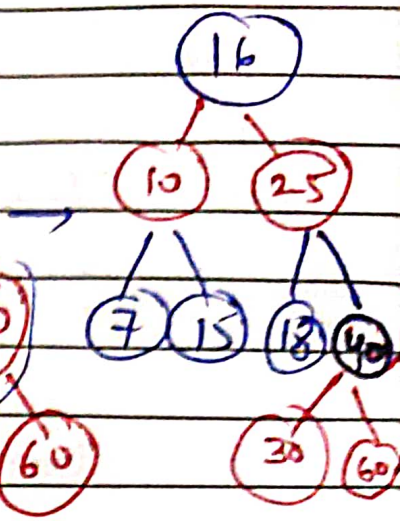
recolor of parent & parent's parent

∴ 16, 10 → recolor

insert 60



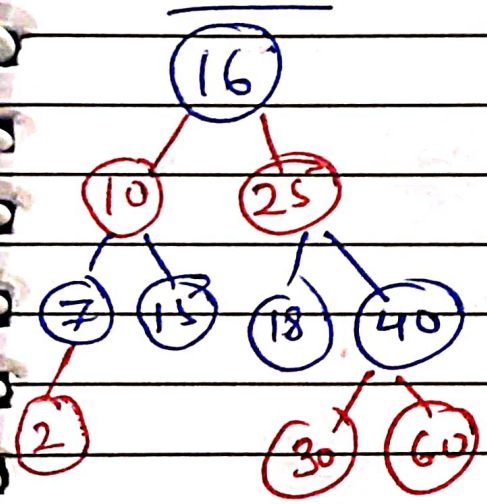




Here parent sibling is NULL.
rotation

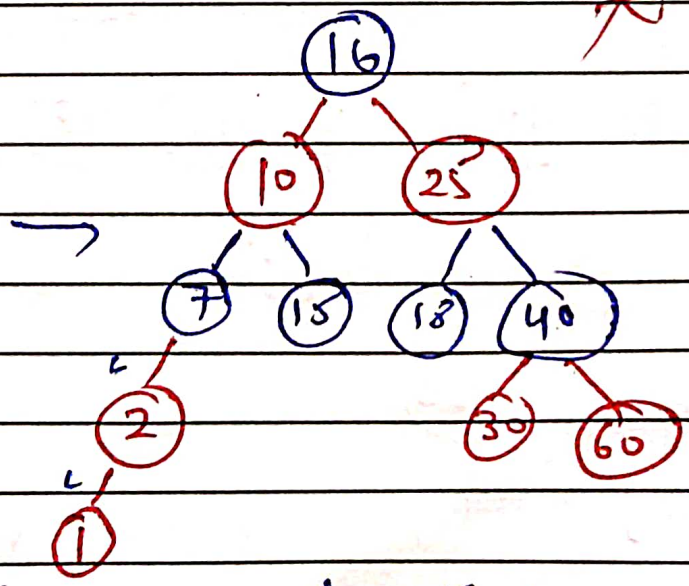
recolor.

insert 2

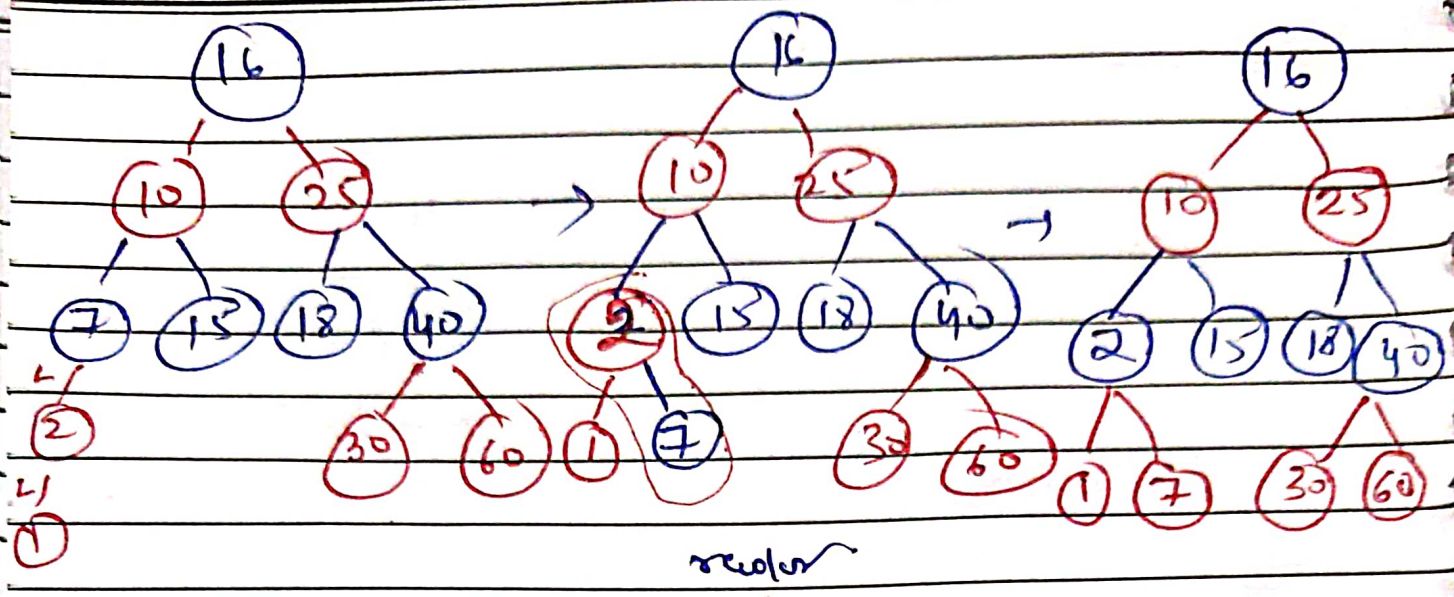


parent of node is black simply exit.

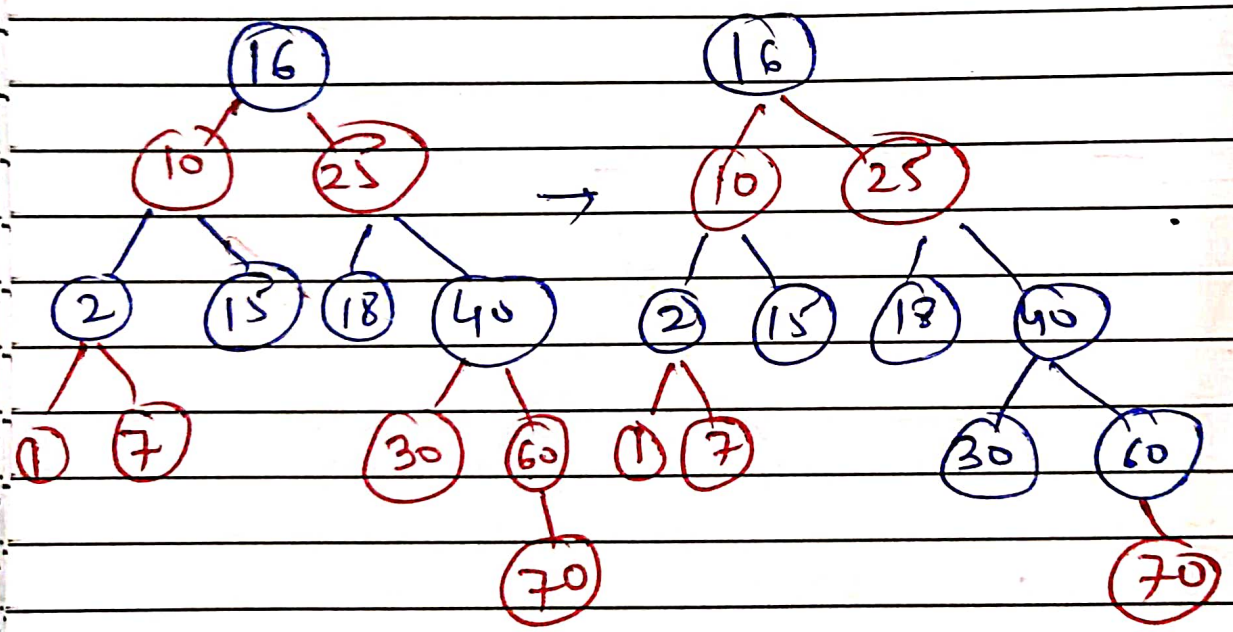
insert 1



parent's sibling is NULL (2)
rotation & recolor

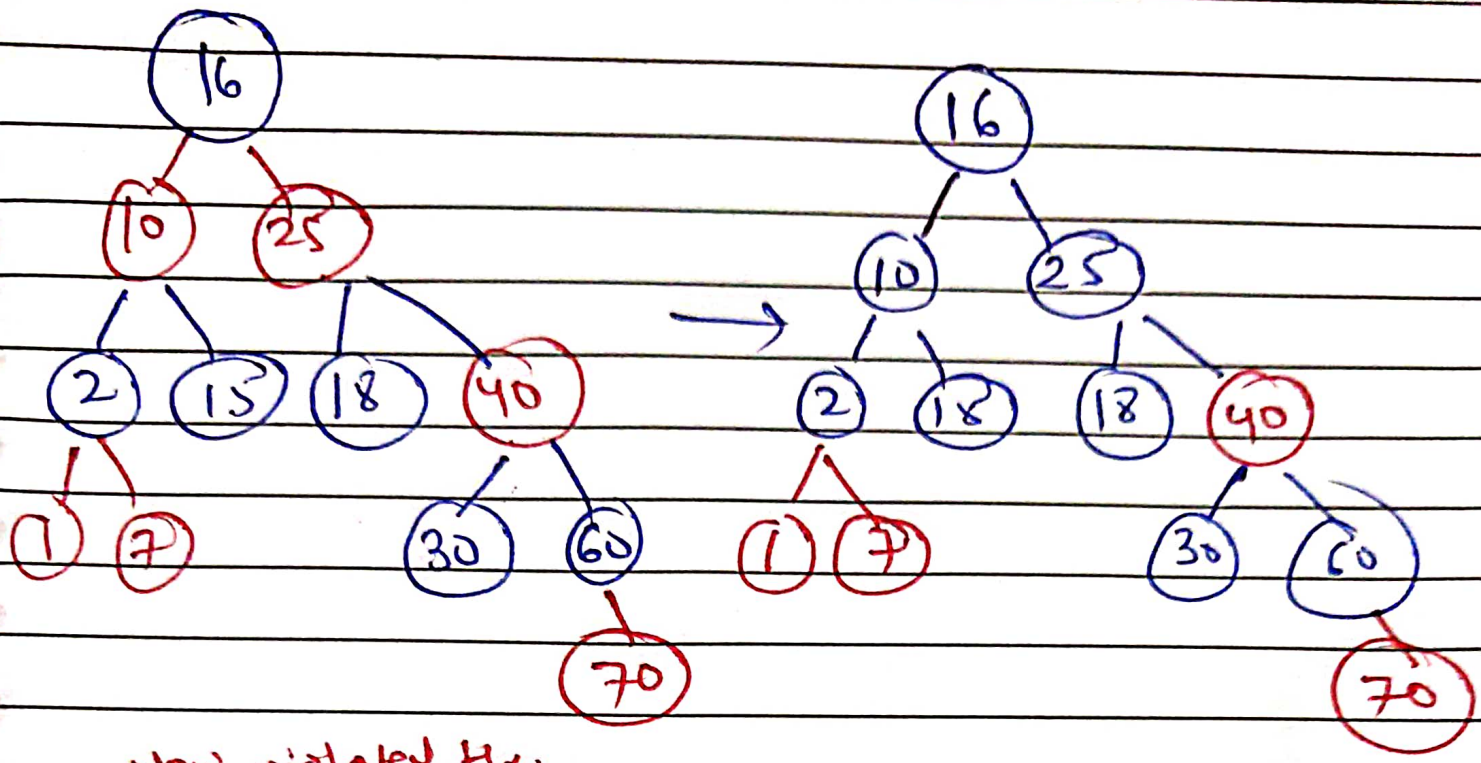


insert 70



Parent's sibling is red
(60) (30)
recolor parent & sibling

parent's parent is not
(60) (40)
root node
recolor



Now violated the property of RB tree then 40 is considered as new node.

check parent's sibling is red (25) (10)

then recolor parent & sibling

Here parent's (25) parent of 40 is (16) root node then no need to check anit.